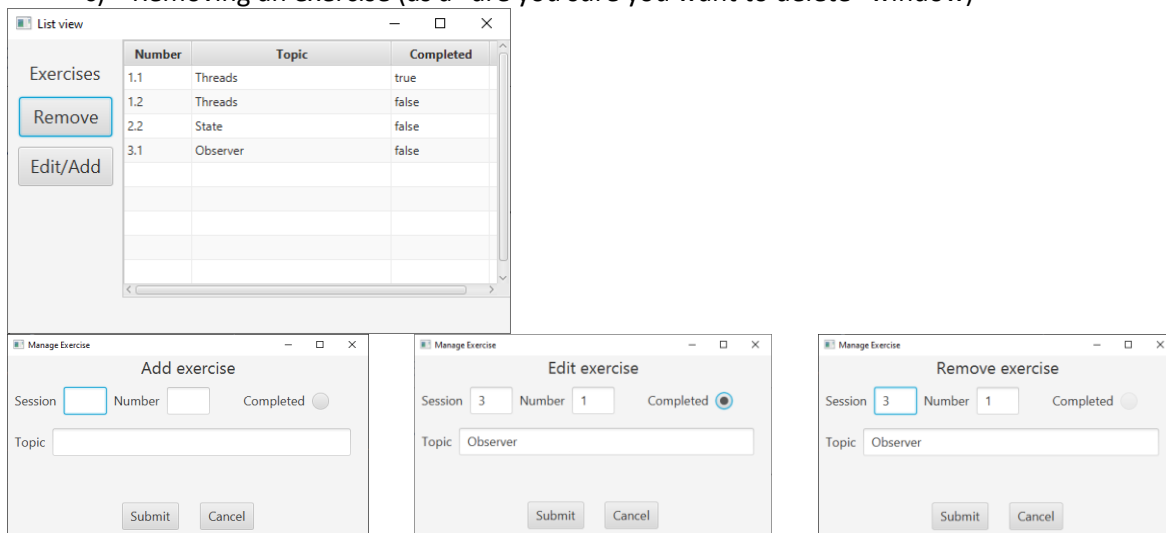


Exercise: MVVM - Exercises

The purpose of this exercise is to create a two-window MVVM application representing SDJ exercises, see below. The two windows are

- Window 1: A list view with a table of exercises as the number, the topic and if completed or not
- Window 2: A manage exercise window, to be used when
 - a) Adding a new exercise
 - b) Editing an existing exercise (e.g. marking it as completed)
 - c) Removing an exercise (as a “are you sure you want to delete” window)



- If no rows are selected, when pressing the Edit/Add button in List view, the Manage exercise window opens with empty text fields – and headerLabel with text “Add exercise”
- If a row is selected, when pressing the Edit/Add button in List view, the Manage exercise window opens with filled out text fields – and headerLabel with text “Edit exercise”
- If a row is selected, when pressing the Remove button in List view, the Manage exercise window opens with filled out text fields and deactivated radiobutton and text fields being un-editable – and headerLabel with text “Remove exercise”
- Observer pattern: The Model fires the following events, with oldVlue being the exercise number (String) and the newValue the Exercise:
 - “Remove”
 - “Add”
 - “Edit”

The full class diagram is given on the next page. A few notes:

- The class SimpleExerciseViewModel represents data in the Table, having properties for Number, Topic and Completed – With types StringProperty, StringProperty and ObjectProperty<Boolean>, respectively
- The Table (in ListExercisesViewController) could be declared this way


```
@FXML private TableView<SimpleExerciseViewModel> exercisesTable;
@FXML private TableColumn<SimpleExerciseViewModel, String> numberColumn;
@FXML private TableColumn<SimpleExerciseViewModel, String> topicColumn;
@FXML private TableColumn<SimpleExerciseViewModel, Boolean> completedColumn;
```
- ... and (after class ListExercisesViewModel is implemented), the Table and its columns could be connected to SimpleExerciseViewModel this way


```
numberColumn.setCellValueFactory(
    cellData -> cellData.getValue().getNumberProperty());
topicColumn.setCellValueFactory(
    cellData -> cellData.getValue().getTopicProperty());
```

```

completedColumn.setCellValueFactory(
    cellData -> cellData.getValue().getCompletedProperty());
exercisesTable.setItems(viewModel.getAll());

```

- Class `ListExercisesViewModel` contains an `ObservableList` of type `ObservableList<SimpleExerciseViewModel>`
- Class `ViewModelState` contains information of the selected row and a boolean if it is remove or not being activated. The `ViewModelState` object is created in the `ViewModelFactory` and passed to the constructors for both `ListExercisesViewmodel` and `ManageExerciseViewModel`. The first one, may set values and the second may get values form this View State object.

Model and FXML files given. Implement the remaining parts of this system.

